Can Frontier LLMs One-Shot Translate Lakota? (Mostly No.)

Evaluating Frontier LLM Translation Capability for Lakota

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No frontier LLM reliably translates Lakota.

Even measuring how badly they fail is hard.

Lakȟótiyapi

the language under test

Why test this?

Lakȟótiyapi (Lakota) is a Siouan language UNESCO classifies as **critically endangered** — fewer than 2,000 first-language speakers, most over 65.

Today's LLMs list translation support across dozens of languages, but providers don't say what they can or can't do per language. A reasonable user can't tell Lakota apart from any of the others on the list — *if it does all these, why not this one?* Direct evaluation is the only way to find out.

Is it just memorized?

The evaluation set comes from a published dictionary, so contamination is the obvious worry. The Lakota source appears verbatim online for ~10% of pairs, but the Lakota and its English reference together for only 0.5%. Scores spread widely (6–61% equivalence), not clustered at ceiling — the signature of partial capability, not recall.

Setup

- **7 models.** Proprietary: Gemini 3.1 Pro, Claude Opus 4.6, Claude Sonnet 4.6, GPT-5.2. Open-weight: DeepSeek-V4-Pro, Qwen3.6-Plus, GLM-5.1.
- **200 sentence pairs** from the New Lakota Dictionary, conversational register, **both directions**.
- **Two conditions:** baseline vs. extended reasoning, where the API permits.
- **Metrics:** chrF++ and BLEU; diacritic-normalized chrF++; an LLM *semantic* judge on the English side.

Why two judges?

chrF++ rewards surface overlap; it cannot tell fluent-but-wrong from correct. Meaning needs a separate check. For Lakota → English the model output *is* English, so judging meaning takes no Lakota at all — that's what made an LLM judge feasible.

To keep one model from flattering its own family, every pair is judged by **two models from different families** — Gemini 3 Flash and the independent Llama-3.3-70B — which agree substantially: Cohen's $\kappa = 0.75$ (0.89 quadratic-weighted).

Standard Lakota Orthography

ȟ ą į ų á ŋ ʼ

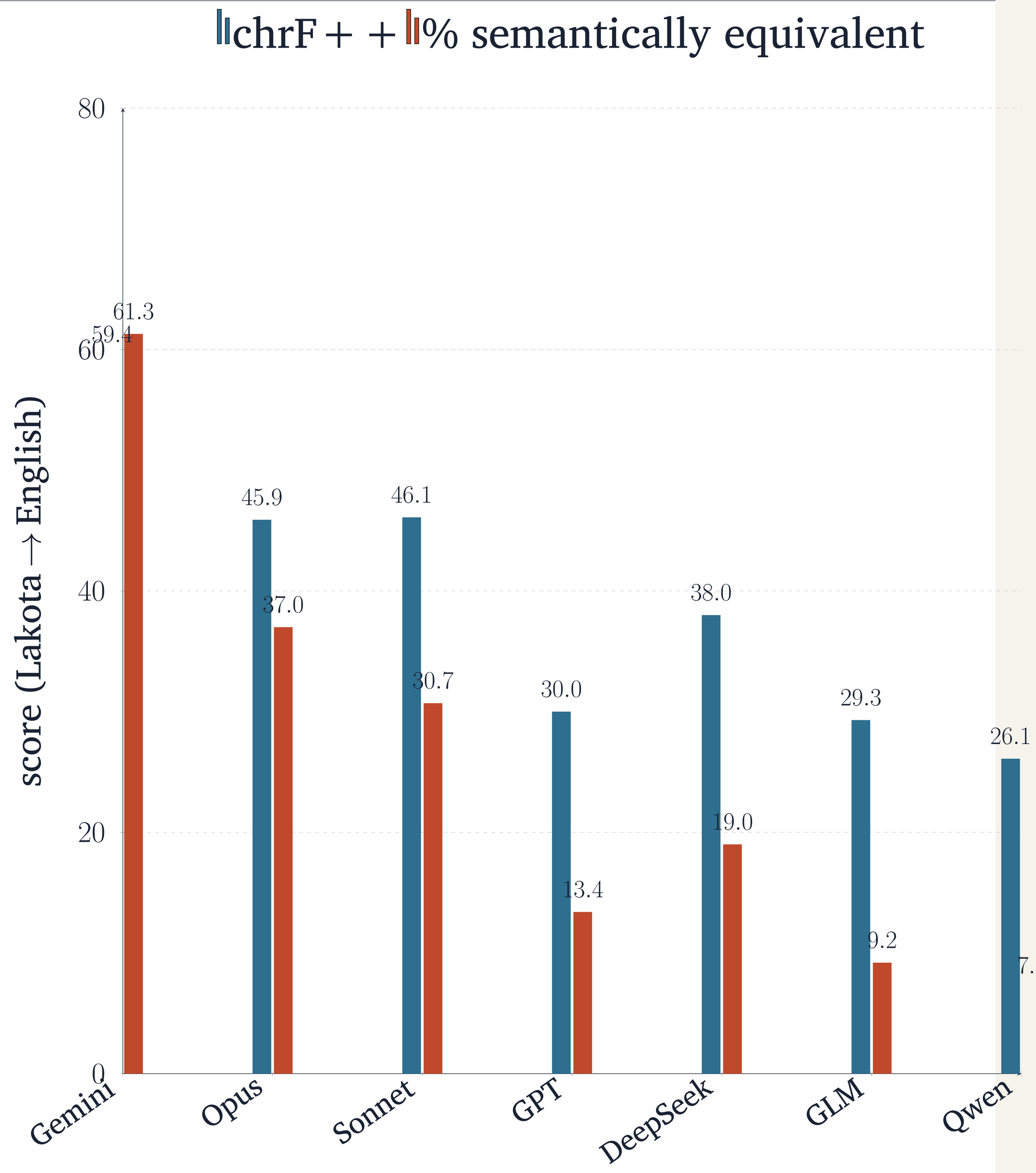
SLO encodes phonemic distinctions with diacritics; stripping them adds **+3.3 to +6.5 chrF++** (E → L) — models get base characters right, marks wrong.

Translation quality (chrF++)

Model	L → E	E → L
Proprietary		
Gemini 3.1 Pro	**59.4**	**42.6**
Claude Opus 4.6	45.9	34.3
Claude Sonnet 4.6	46.1	27.0
GPT-5.2	30.0	23.6
Open-weight		
DeepSeek-V4-Pro	38.0	29.2
GLM-5.1	29.3	14.6
Qwen3.6-Plus	26.1	18.4

Best condition per model. Every model scores higher *out of* Lakota than *into* it; no open-weight model approaches Gemini.

Surface metrics overstate capability



chrF++ stays in the 30s–40s even where genuine semantic equivalence collapses below 15%. The two metrics agree for the strongest model and diverge sharply for the rest — so chrF++ alone is misleading for low-resource translation.

Semantic equivalence (L → E)

Model	Equiv.	Partial	Not
Gemini 3.1 Pro	**61.3**	23.1	15.6
Claude Opus 4.6	37.0	21.0	42.0
Claude Sonnet 4.6	30.7	23.8	45.5
DeepSeek-V4-Pro	19.0	14.0	67.0
GPT-5.2	13.4	9.8	76.8
GLM-5.1	9.2	6.5	84.2
Qwen3.6-Plus	7.1	7.6	85.4

% of judged pairs, reasoning condition.

How models fail

Reference	Model output	chrF++
We live in a small trailer house with no electricity.	"small *sacred* tipi"	18.6
Make coffee (*Wakȟálya yo*).	"Sanctify it!"	2.9
Did my daughter call?	"Did my younger brother come?"	27.6
Do you have a car?	"Do you have a lighter?"	65.9
I want to talk to you.	"I will be very happy."	6.5

Idioms and coined terms are translated *element by element*: *Wakȟálya yo* ("make coffee," literally "make it boil") becomes "sanctify it." The one-word near-miss (65.9) outscores output unrelated to the source — surface overlap and meaning preservation diverge.

Reasoning buys calibration, not capability

For the open-weight models, extended reasoning leaves translation quality and meaning *flat* but raises **refusals**:

- Qwen3.6-Plus, E → L: **0** → **36** refusals (of 200) when reasoning is enabled.
- GLM-5.1, L → E: **1** → **16**.

Reasoning surfaces the limitation rather than overcoming it. On a language this far outside the training distribution, the binding constraint is **knowledge, not inference** — reasoning's contribution is calibration, not capability.

Takeaways

- No frontier LLM reliably translates Lakota in either direction (as of 2026).
- **chrF++ overstates** low-resource capability — pair it with an LLM semantic judge.
- Open-weight models do not close the gap; reasoning changes *refusals*, not quality.

Origin

This evaluation began as a response to a provocation from Benjamin Bratton (UCSD): *"just vibe-code a Lakota translation app over the weekend."* We built the evaluation that would test whether such an app could exist.

Data & code

github.com/robotson/lakota-translation-benchmark

Evaluation corpus withheld (community data); the repo holds scripts, aggregate results, and the paper.